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SYSTEMATIC REVIEW

Cryoneurolysis for phantom limb pain: a systematic review

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ABSTRACT

Aims: Phantom limb pain (PLP) is a painful sensation occurring in patients around the site of an amputation. The aim of this systematic review is to evaluate the efficacy of cryoneurolysis in the management of phantom limb pain.

Materials and methods: A systematic review was performed according to the PRISMA 2020 guidelines. An initial search yielded 200 articles from four major scientific databases (PubMed, Embase, Cochrane Library, WebOfScience). Five articles met inclusion criteria, four of which underwent additional pooled statistical analysis.

Results: Pooled analysis of the included trials revealed a cumulative Cohen's d effect size of 1.55 (95% CI [0.24, 2.87]; $p = 0.02$; $z = 2.32$) for the reduction of pain on a 10-point pain scale following cryoneurolysis intervention. The remaining article that did not meet inclusion criteria for statistical analysis was a case report that reported a reduction in pain from 9/10 to 1/10 one week following intervention.

Conclusions: The large effect size demonstrated a statistically and clinically significant improvement in patient-reported pain. Additionally, patients may be able to reduce their amount of pharmaceutical pain management with successful cryoneurolysis treatment. However, these findings are limited by the small sample size and high heterogeneity between studies. Further high-quality studies should be performed to corroborate these findings.

Protocol registration: www.crd.york.ac.uk/prospero identifier is CRD42024543085.

FREEZING NERVES TO TREAT PHANTOM LIMB PAIN: A REVIEW OF CURRENT RESEARCH - PLAIN LANGUAGE SUMMARY

Phantom limb pain (PLP) is a challenging condition where individuals feel pain in the area where a limb has been amputated. This systematic review examined the use of cryoneurolysis, a procedure that freezes nerves to reduce pain, to manage PLP.

Researchers followed PRISMA 2020 guidelines for the review, analyzing studies from major databases, including PubMed and Cochrane Library. Out of 200 initial studies, five met the criteria for inclusion, with four providing data for a combined statistical analysis. The fifth study, a case report, was analyzed separately.

The analysis showed a large positive effect of cryoneurolysis on reducing PLP. On a 10-point pain scale, patients reported significant pain relief, with the pooled data showing a strong reduction in pain scores. In the case report, the patient's pain dropped from a severe 9/10 to a mild 1/10 just one week after the treatment.

The findings suggest that cryoneurolysis is an effective option for reducing PLP and may also allow patients to use less medication to manage their pain. However, the conclusions are limited by the small number of studies and differences in study methods. More high-quality research is needed to confirm these promising results.

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KEYWORDS
Cryoneurolysis; phantom limb pain; amputation; cryoneuroablation; cryoanalgesia

1. Introduction

Phantom limb pain (PLP) is a phenomenon that describes painful sensations perceived in areas of the body that have been amputated or surgically removed [1]. PLP can occur in up to 66.55% of patients in developed countries [2] and is typically described as a sharp, burning, cramping, or paresthesias – different than postoperative pain and residual limb pain [3]. Phantom limb pain occurs more often in the upper extremities than in the lower extremities [4]. Moreover, phantom limb pain has a predilection to affect adults juxtaposed to children, with symptoms either starting immediately after surgery or having delayed onset [5]. As such, PLP can cause significant debility, distress, and reduction to a patient's overall quality of life [6].

There are currently several theories that seek to explain the cause of this condition. One theory [7] states that prior to amputation, there is consistent communication between the limbs and higher cortical areas. However, after an amputation, it is thought

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Article highlights

Phantom limb pain and cryoneurolysis
- Phantom limb pain, a perceived painful sensation in areas of the body that have been amputated or surgically removed, has an ambiguous and often multimodal approach to treatment
- Cryoneurolysis is a procedure that induces nerve degeneration and subsequent pain relief with axon regeneration that may be used to treat PLP

Systematic review and pooled analysis
- A systematic review following PRISMA 2020 guidelines and pooled analysis of pain outcomes after cryoneurolysis for phantom limb pain was conducted

Cryoneurolysis may be effective in reducing pain associated with PLP
- Pooled analysis of four included studies ($n = 104$) demonstrated a large and statistically significant pain reduction following cryoneurolysis
- Notably, the largest included study showed no statistically significant difference in pain reduction between cryoneurolysis and sham treatment, highlighting the variability in outcomes across studies

Uses of cryoneurolysis in the management of PLP
- Cryoneurolysis shows significant potential for reducing PLP and may offer an alternative or adjunct to standard treatments, potentially decreasing opioid dependence in PLP management

that there is an apparent mismatch in communication between the nerves of the amputated limb and the brain. Essentially, the missing limb's region in the primary somatosensory cortex undergoes "brain reorganization" due to a lack of sensory input. Additionally, chronic pain in the residual limb can lead to a maladaptive rearrangement of nociceptive neural pathways at the dorsal horn of the spinal cord due to overstimulation, resulting in increased spontaneous activity, decreased activation threshold, and increased response to suprathreshold stimulus [8]. Another model for PLP development is the neuroma model, from which PLP is caused by an ectopic hyperexcitable area on the severed nerves of the residual limb [9]. However, not every patient who suffers from PLP develops a neuroma in their residual limb.

Moreover, similar to other chronic pain conditions, psychological factors are important components of pain perception in PLP [10]. Chronic stress, anxiety, and depression are believed to be contributing factors to the development of PLP through the diathesis-stress model for chronic pain [11]. This can lead to a positive-feedback loop, as the debilitating pain can reduce the individual's capacity for self-care, independence, and lead to depression and feelings of hopelessness [12].

Treating PLP is often a significant challenge to clinicians. This pathology is often multifaceted, with emotional, cognitive, and sensory factors contributing to the patient's disease state [13]. Synergizing with this, effective treatment modalities are limited. However, mirror therapy, often performed by physical therapists, has been found to mitigate pain and improve morale in patients with PLP [14]. Although still controversial, there is support for mirror therapy as an effective treatment for those suffering from chronic PLP symptoms [15]. Conversely, phantom exercises, aka phantom motor execution, which involves coupling imagining phantom limb movements with certain physical movements, have been shown to have better pain management than mirror therapy and physiotherapy [16]. Additionally, patients with PLP are often treated pharmacologically with antidepressants, anticonvulsants, or opioids to provide alleviation [1]. Less frequent interventions include neuromodulation strategies such as transcutaneous electrical nerve stimulation and transcranial magnetic stimulation [17].

A potential intervention for PLP includes cryoneuroablation of the severed nerve or formed neuroma. Cryoneuroablation is also referred to as cryoanalgesia, nerve cryoablation, or cryoneurolysis [18]. Nerve cryoablation works by induction of Wallerian degeneration, at −20°C to −100°C, distal to the induced lesion, leading to sustained pain relief, while the axon regenerates [19]. Cryoneurolysis has been utilized in the treatment of other pain pathologies, such as chronic knee osteoarthritis and intercostal nerve ablations [20–22]. However, the consensus on cryoneurolysis' application for patients afflicted with phantom limb pain has yet to be discerned.

Cryoneurolysis may be a potential treatment for PLP. To date, very few studies have investigated its efficacy. The objective of this systematic review is to investigate the potential application of nerve cryoablation-induced analgesia to the residual limb nerve or neuroma to relieve PLP.

2. Methods

A rigorous systematic review utilizing the PRISMA 2020 guidelines [23] and pooled analysis on pain outcomes following cryoneurolysis of severed nerve or associated neuroma for phantom limb pain were conducted. This study was registered on Prospero (Reference: CRD42024543085).

2.1. Inclusion criteria

Articles must contain primary clinical data on pain outcomes following cryoneurolysis of severed nerves or associated neuromas in patients with phantom limb pain to be included in the study. The study designs included in our statistical analysis included randomized controlled trials, single-armed trials, case series, case reports with multiple patients, and retrospective cohort studies. Single-patient case reports are reported in narrative format, but not included in statistical analysis, as their lack of standard deviations and variance inhibits effect size calculations. The primary outcome is a change in pain scores following cryoablation of affected nerves.

2.2. Exclusion criteria

Articles were excluded from analysis if full-text was not available after extensive searching and inter-library loan requesting, the statistics were unable to be pooled or modified to be included in the pooling, the article did not report pain scores, or the study reported potentially confounding residual-limb /stump pain. Articles were not excluded based on country of origin, date of publication, or language of origin so long as the researchers could accurately translate the language. Articles without a nerve or neuroma cryoablation groups were excluded. Studies in progress were not excluded so long as there was a preliminary report of their resultant data. Articles

with potentially confounding concomitant treatments were excluded from analysis. Articles discussing the prevention of PLP rather than the active treatment of PLP were excluded from analysis.

2.3. Information sources and search strategy

PubMed, Embase, Cochrane Library, and WebOfScience were queried on 5/15/24 for this systematic review and pooled analysis. Scopus was not queried due to persistent technical difficulties. The search string employed in database querying was developed using MeSH key terms and major keywords of the therapeutic technique and condition of interest. The search string was: ("Cryoneurolysis" OR "cryoablation" OR "cryoanalgesia" OR "PerCryo" OR "Iovera" OR "Cryo") AND ("Phantom Limb Pain" OR "Phantom limb" OR "Phantom limb pains" OR "Phantom pain" OR "Phantom pains" OR "Pseudomelia" OR "Pseudomelias" OR "Phantom sensation" OR "Phantom sensations" OR "amputation" OR "amputee" OR "limb loss" OR "postamputation"). Following database querying, exhaustive manual bibliographic and citation searching were performed to ensure no articles were missed.

2.4. Study selection

Articles resulting from the initial database query were subjected to duplicate detection and removal using Rayyan.ai. Articles were then sorted by two independent reviewers (BG and JD), with a third being brought in to break ties (UC). At this stage, articles were assessed for adherence to inclusion and exclusion criteria. After extensive review, a full-text appraisal of the remaining articles was conducted utilizing the same approach. If an article was deemed to meet all necessary criteria after full-text investigation, and the data was deemed to be relevant, data was then extracted to an external data collection form and uploaded to the Statistical Package for Social Science (SPSS) version 21 (Armonk, NY: IBM Corp) for analysis.

2.5. Data collection

Data from included articles was extracted by two reviewers onto an external data extraction sheet. The outcome of interest was reduction in pain status post cryoneurolysis of severed nerve or neuroma in patients suffering from phantom limb pain. Data points were then uploaded to SPSS for pooled analysis.

2.6. Certainty of evidence and risk of bias assessment

Articles included in analysis were then scored for certainty of evidence utilizing the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) criteria [24]. The GRADE criteria and explanation for any decisions to downgrade the quality of evidence are detailed in the summary of findings chart, which was adopted from GRADEpro GDT [25] (Table 1).

Bias in included articles was assessed using metrics designed for their specific study type. ROBINS-I was employed to encompass the included study types [26]. A description of the domains and the respective determination for each study is included in Figures 2 and 3, which were generated using robvis [27].

3. Results

The database search yielded 200 articles. Following duplicate detection, 138 articles remained. These articles underwent title and abstract appraisal with only 12 articles meeting inclusion and exclusion criteria for full-text appraisal. Following a full-text review, four articles [28–31] were included in the statistical analysis (Figure 1).

3.1. Excluded articles after full-text appraisal

Three articles did not have available full text following exhaustive searching. Five articles did not report usable statistics. Two articles were the wrong publication type. Of these, one article was on the prevention of PLP and not active treatment and was subsequently excluded. If the statistics were unclear or unusable, the corresponding author was contacted with a request for raw data or clarification. Corresponding authors were given 1 month to respond and were excluded if no response was received.

3.2. Effect of intervention

To determine which specific model to use (fixed/common-effects or random-effects), an initial heterogeneity test (Higgins) was utilized to discern I^2, which found that $I^2 = 89.5\%$. As such, random-effect models were employed in pooled analysis to control for potential confounding. Due to the high heterogeneity, a meta-analysis was not performed.

As of 5/15/2024, 104 patients had undergone cryoneurolysis of a severed nerve or resultant neuroma for phantom limb pain ($n = 104$). When comparing the post-intervention pain levels of PLP patients, a significant effect size was found to reduce pain

Table 1. Summary of findings of GRADE assessment of certainty of evidence.

Participants (Studies)	Risk of Bias	Inconsistency	Indirectness	Imprecision	Other Considerations	Overall Certainty of Evidence
Effect of nerve cryoablation on phantom limb pain						
104 (1 prospective cohort study, 2 retrospective cohort studies,	Not serious	Not serious	Serious	Serious	Undetected	Low
Explanation		High heterogeneity not factored due to use of pooled analysis over meta-analysis	Analysis of indirect comparisons; different methods of administering cryoablation; inconsistent "types" of phantom limb	Small sample size		

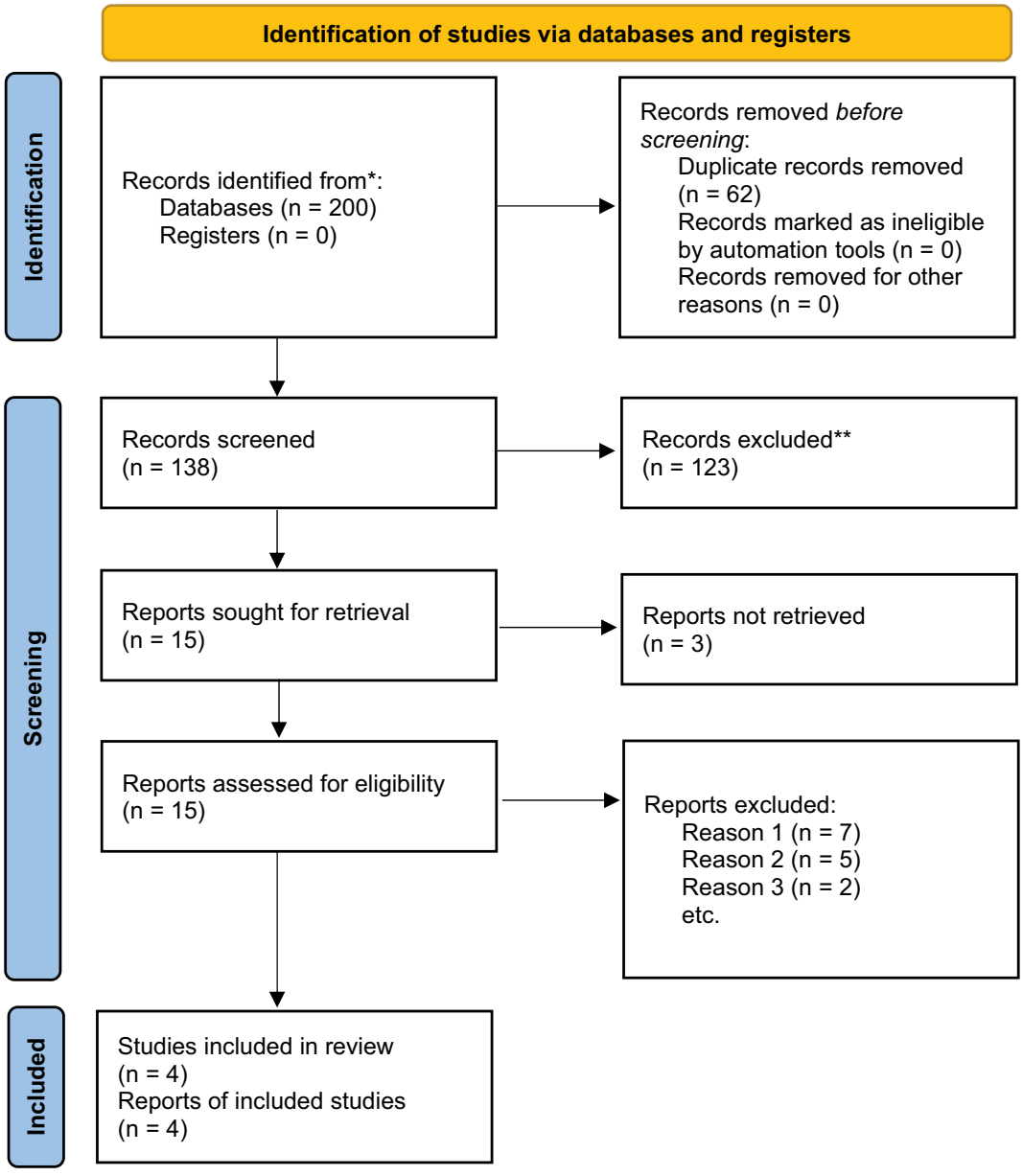

Figure 1. PRISMA 2020 flowsheet of included articles.

levels. Of the studies included in the quantitative statistical analysis, the pooled-effect size analysis yielded a Cohen's d = 1.55 (95% CI: 0.24, 2.87; $p = 0.02$). All included studies report utilizing a 10-point pain scale such as Visual Analog Score (VAS), Numerical Rating Score (NRS), or a similar measure. It is important to note that the largest and strongest multi-center study to date [31] does not report a significant difference in pain scores following cryoablation of affected nerves and neuromas (d = 2.72; $p = 0.107$) compared to all other included articles.

3.3. Narrative review

Moesker et al. [28] utilized cryoanalgesia at a location determined by a diagnostic nerve block in five patients suffering from phantom limb pain. Over 2.5 years, three patients reported a decrease in pain between 90% and 100%, one reported a 40% decrease in pain, and the last one reported a 20% decrease in pain. Additionally, in one patient, morphine mg/24 h was halved, gabapentin reduced by 75%, and amitriptyline reduced from 45 to 35 mg/24 h. Another patient stopped their 1800 mg/24 h of gabapentin entirely. The authors concluded that although cryoanalgesia of the peripheral pain locus may help provide long-term pain relief, there are still central components to the pain.[28]

Prologo et al. [29] treated 21 patients with refractory phantom limb pain with image-guided percutaneous cryoneurolysis. At 180-day follow-up, pain scores on the VAS decreased from 6.2 to 2.0 (95% confidence interval 2.8, 5.6; $p < 0.0001$). Furthermore, patients were administered a modified Roland Morris Disability Questionnaire, which revealed a decrease of 11.3 to 3.3 at 45-day follow-up (95% confidence interval 5.8, 10.3; $p < .0001$).

Von Falck et al. [30] utilized ultrasound-guided cryoablation of a previously identified painful stump neuroma on seven patients experiencing phantom limb pain. Using baseline VAS pain score

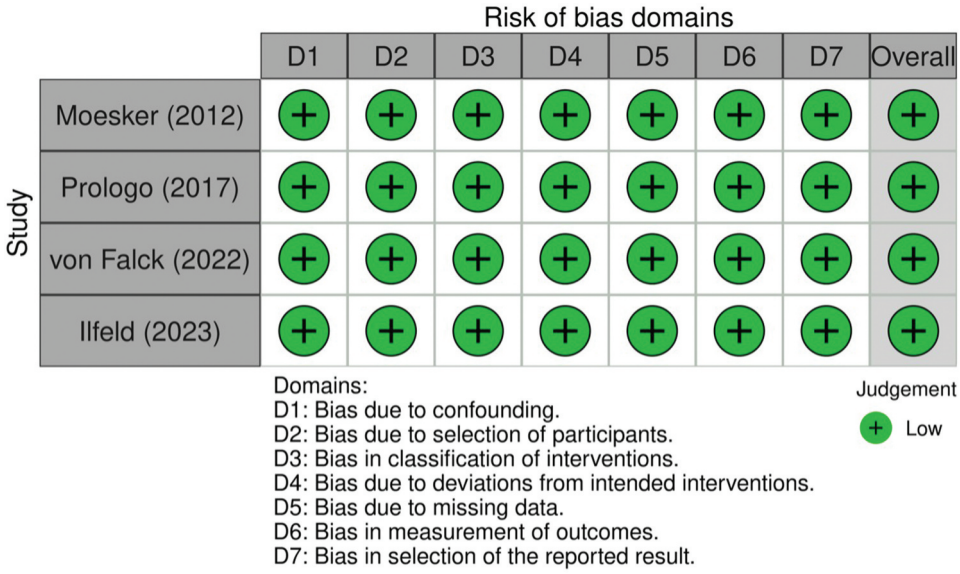

Figure 2. ROBINS-I traffic-light plot.

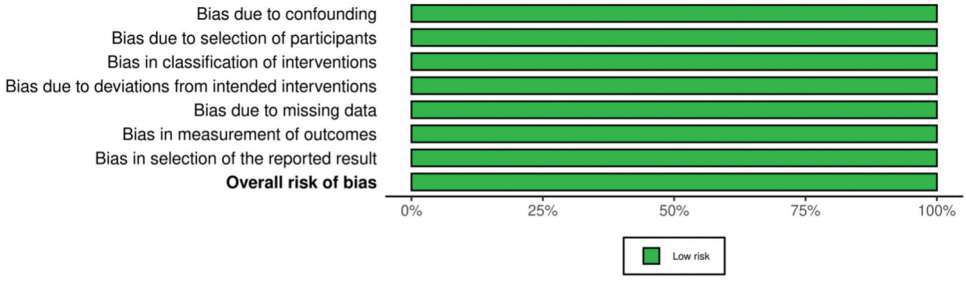

Figure 3. ROBINS-I summary plot.

as a comparison (8.3), the authors appreciated a statistically significant reduction in pain at 1 day (4), 1 week (2.1), and the last follow-up (3) ($p < 0.05$). Only one patient was not satisfied with the treatment; the patient reported a 30% reduction in pain after cryoablation and was not willing to undergo re-ablation.

Ilfeld et al. [31] followed the pain scores for an ultrasound-guided percutaneous cryoneurolysis group ($n = 71$) and a sham treatment group ($n = 73$). At the 4-month follow-up, pain scores were reduced by 0.5 [−0.5, 3.0] in the treatment group and 0 [0, 3] in the sham group. The estimated difference was (95% CI) of −0.1 (−1.0 to 0.7), $p = 0.759$. Per these results, the cryoneurolysis failed to statistically significantly reduce the pain in comparison to sham treatment. This is especially significant as this study was the largest included in the review.

3.4. Certainty of evidence and risk of bias assessment

GRADE analysis revealed an overall low certainty of evidence. Indirect comparisons and a small sample size lowered the certainty of evidence. Other factors that are considered for upgrading the quality of evidence (i.e., large effect size, dose–response gradient, and plausible confounders that would have reduced effect size) were not applicable (Table 1).

The risk of bias assessment revealed a low risk of bias in all four included studies due to negligible concerns with risk of bias across the seven domains (Figures 2 and 3).

4. Discussion

Cryoneurolysis demonstrates promise as an alternative intervention for phantom limb pain due to its ability to reduce nociceptive transmission within the targeted nerve or neuromas. Despite the sizable population affected by PLP, cryoneurolysis has been minimally utilized due to the novelty of intervention, prompting a systematic review of the literature. This systematic review aimed to determine the effect of cryoneurolysis as an intervention for managing phantom limb pain by analyzing pain scores post-intervention to the baseline pain scores. The results show a large treatment effect for the reduction of pain, with an overall pooled effect size of 1.55.

4.1. Implications

4.1.1. Reduction of pharmaceuticals

PLP is a difficult condition to treat, often requiring a multidisciplinary approach [32]. Given its favorable results, cryoneurolysis provides a potential alternative to refractory PLP or a beneficial adjunct to conservative or standard therapy. One

potential benefit is an opportunity to reduce medication use in PLP, particularly opioids. As opiates and other pharmaceutical options are often first-line for PLP, there is a risk for dependence and abuse [33]. Cryoneurolysis has been shown to decrease opioid use in various procedures, most notably in thoracotomy. A retrospective chart review by Chidiac et al. [33] demonstrated that patients treated with cryoablation during thoracotomy for pulmonary metastases received a lower total dose of opioids compared to non-treated patients (2.2 oral morphine equivalent per kilogram versus 14.4, $p = 0.004$). Additionally, another retrospective chart review by Dasa et al. [34] showed that patients with preoperative cryoneurolysis treatment before total knee arthroplasty required 45% fewer opioids than non-treated patients. With Moesker et al. [28] demonstrating decreased use of analgesics in its results, cryoneurolysis may be an option for reducing opioid use in PLP.

4.1.2. Effectiveness in other conditions

The success of cryoneurolysis is not limited to PLP [18]. Many other conditions treated with cryoneurolysis have shown promising pain reduction results in acute and chronic settings. Successfully treated conditions include management of joint pain, temporomandibular disorders, headaches, and cancer-related pain [35]. For example, in a double-blind RCT of cryoneurolysis in knee osteoarthritis [21], cryoneurolysis showed significantly decreased knee pain and improved symptoms compared to sham treatment. Similarly, in a prospective clinical trial of 46 patients with degenerative lumbar spine disease [36], cryoneurolysis of lumbar facet joints showed significant pain reduction with 72% of patients pain-free or improved low back pain at 6 weeks and 57% at 1 year. This was shown with a decrease in VAS from 7.7 preoperatively to 3.2 at 6 weeks and 4.2 at 12 months. Another promising application for cryoneurolysis is in the management post-thoracotomy pain. This pain often arises in thoracotomy-related procedures such as mastectomies and rib fractures. Cryoneurolysis in these conditions targets the intercostal nerves. Many studies have shown significant improvement in pain reduction following thoracotomy [22]. A recent case report [37] demonstrated success in ultrasound-guided percutaneous cryoneurolysis for post-thoracotomy pain with an immediate 75% pain reduction and 50% pain reduction at 14 weeks. Conversely, other studies have reported a failure of pain reduction and potential neuroma formation, making it a more controversial treatment option for post-thoracotomy pain [38]. Further research is needed to evaluate the current efficacy in post-thoracotomy pain.

4.2. Limitations and future directions

This study had multiple limitations. Despite the reduction in pain scores for patients with PLP following cryoneurolysis, this study was limited by small sample sizes, high heterogeneity between studies, and lack of strong certainty of evidence. Only one randomized-controlled trial has been conducted to date investigating PLP, and it has failed to show significance with its primary endpoints. Additionally, methods between cryoneurolysis treatments (such as temperature selection) are not uniform as there is yet to be an accepted standard protocol due to a lack of research. Some studies used diagnostic blocks prior to cryoneurolysis, while others did not. Further studies should investigate if a prior positive diagnostic block increases the efficacy through accuracy.

5. Conclusion

Cryoneurolysis is a novel treatment modality that shows some promise for use in treating phantom limb pain. The present study depicts a large effect size for the use of cryoneurolysis in PLP; however, the largest randomized-controlled trial to date failed to show significance. As a result, it is imperative that further larger studies are conducted to investigate the potential benefits of cryo nerve-ablation in PLP management. Moreover, a consensus should be reached regarding the benefit of pre-cryoanalgesia diagnostic blocks in PLP-centric cryoneurolysis.

Author contributions

Hanna Brancaccio: Data Curation, Writing – Review & Editing, Visualization **Brandon Goodwin**: Conceptualization, Formal Analysis, Data Curation, Writing – Original Draft, Project administration **John DesRochers**: Writing – Original Draft **Alec Birnbaum**: Writing – Original Draft **Uzay Cagatay**: Writing – Original Draft **Bianna Koutsenko**: Writing – Original Draft **Connor Flatley**: Writing – Review & Editing **Gilbert Siu**: Supervision

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